There exist irrational numbers x and y such that

x^y is rational.

→ We are not asked to find "all" such pairs

→ to give a specific numeric value right away.

→ We are only asked to prove existence.

We already know $\sqrt{2}$ is irrational.

<u>Step #1</u>: Any real number must be either

– rational or – irrational. There is No third option (2 exhaustive cases)

So for $\sqrt{2}^{\sqrt{2}}$, exactly one of these is true.

<u>Case A</u>: $\sqrt{2}^{\sqrt{2}}$ is rational

Then choose: $x = \sqrt{2}$ (irrational)

$y = \sqrt{2}$ (irrational)

We get $x^y = \sqrt{2}^{\sqrt{2}}$ (rational)

Existence proven.

<u>Case B</u>: $\sqrt{2}^{\sqrt{2}}$ is irrational

Then choose: $x = \sqrt{2}^{\sqrt{2}}$ irrational, by assumption

$y = \sqrt{2}$ irrational

We get $x^y = \left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} = \sqrt{2}^{(\sqrt{2}\cdot\sqrt{2})} = \sqrt{2}^2 = 2$

and 2 is rational.

Existence proven again.

No matter which case is true, we always end up with

irrational x
irrational y
rational x^y

So at least one of the 2 pairs must work.

This is called nonconstructive because

- We proved that a solution must exist
- But we did not identify which pair actually works.

<u>In logic terms:</u> (Pair A works) $\vee$ (Pair B works)

<u>Proof constructive:</u> must

1. Explicitly give a specific pair (x, y)
2. Prove: x is irrational
   y is irrational
   x^y is irrational
3. Without using a "case split" like "either rational or irrational!

example $x = \sqrt{2}$, $y = \log_{\sqrt{2}}(9)$

Then

$$x^y = (\sqrt{2})^{\log_{\sqrt{2}}(9)} = 9$$

$$\log_{\sqrt{2}} 9 = \frac{\ln 9}{\ln \sqrt{2}} = \frac{\ln 9}{\frac{1}{2}\ln 2} = \frac{2\ln 9}{\ln 2} = \frac{\ln 81}{\ln 2} = \log_2 81 = 6.339850002 8\ldots$$

irrational number